



☯ Active Learning Guide

Simulator: Regression.

This simulator allows users to explore a regression model of the form:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 A + \beta_4 B + \beta_5 AX + \beta_6 BX + \beta_7 AB + \varepsilon$$

where X is a quantitative variable and A and B are binary independent variables.

Users can adjust the regression parameters (β coefficients) and visually observe how the simulated data changes.

It is also possible to add Gaussian noise ε with mean 0 and modify the number of simulated points using a slider, making it an effective tool for understanding the effects of model parameters and randomness on regression outcomes.